

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID - SIH26002

Problem Statement Title - AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

Theme - Smart Automation

PS Category - Software

Team ID - [REGISTERED TEAM ID]

Team Name - NER-AI Logistics

WORKING PROTOTYPE

RASTA-AI

Route Accessibility, Safety,
Terrain & Analytics

Resilient logistics intelligence for
North East India - designed to scale
across India.

NER-FIRST PILOT | PAN-INDIA ARCHITECTURE



RASTA-AI: Resilient Logistics Intelligence Platform



PROBLEM WE SOLVE

- Monsoon, landslides and road disruptions can invalidate a planned route.
- Weak connectivity makes phone-only navigation fragile.
- Generic fleet tools track location, but do not explain operational route risk.



PROPOSED SOLUTION

- Manager + Driver applications on one shared backend.
- Real routing + weather-aware Route Risk V1.
- Explainable route recommendation and human-approved rerouting.
- Live GPS freshness + offline-resilience foundations.



WHAT MAKES IT UNIQUE

- NER Road Memory concept: recurrence, closure, repair evidence and verified-open status.
- Evidence-first design: no fake route, no fake ETA, no fake AI.
- Deterministic intelligence today; ML only after labelled ground truth exists.

PROTOTYPE TODAY

Turn-by-turn navigation

Live GPS + freshness

Real route + backup

Weather-aware risk

Reviewer-authorised selection

Human-approved reroute

Driver progress

en / hi / as catalogue

1. CLIENTS

 **Manager Web**
React + TypeScript **Driver App**
React Native + Expo

2. CORE PLATFORM

FastAPI / Python Services



Auth + RBAC

Trip State +
DispatchGPS Ingest +
FreshnessRouting +
RecommendationWeather + Route
RiskReroute + Offline
Package

3. DATA / PROVIDERS

 **OSRM**
Routes, alternatives + turn steps **Open-Meteo**
Weather along corridor **Future: GSI + road authorities**
Landslide / reopen evidence

PostgreSQL + PostGIS (Supabase target; demo on isolated local cluster) | trip_routes + maneuvers | gps_points | assignments | audit_logs

WORKING METHODOLOGY

PLAN

SCORE

DISPATCH

TRACK

REASSESS

HUMAN REROUTE

DELIVER

**WHY IT IS FEASIBLE**

- Software-first MVP: phone GPS + existing web/mobile clients.
- Provider abstraction avoids hard vendor lock-in.
- PostGIS fits routes, tracks and future hazard layers.
- Deterministic risk/recommendation works before ML.
- Existing prototype already exercises the full trip lifecycle.

**CHALLENGES / RISKS**

- External routing/weather services have uptime limits.
- Road-reopen truth is fragmented across authorities.
- Dead zones require legal offline map packaging.
- Risk thresholds are not yet statistically calibrated.
- Physical Android + language quality still require field validation.

**MITIGATION STRATEGY**

- Graceful provider failure + explicit stale / unavailable states.
- Human-in-the-loop reroute; no silent route changes.
- CLOSED → REPAIR REPORTED → VERIFY → VERIFIED OPEN.
- Offline package + store-and-forward telemetry design.
- Separate deterministic logic from future ML; keep audit trail.

FUTURE UPGRADE ROADMAP**PHASE A • ROAD MEMORY**

Apply reviewed schema → ingest GSI / authority evidence → recurrence + verified-open history.

**PHASE B • OFFLINE & SAFETY**

Licensed offline basemap + durable queue + TTS + SOS + optional truck GNSS backup.

**PHASE C • PREDICTIVE + INDIA SCALE**

Labelled dataset → calibrated ML risk model → truck constraints / terrain → state-specific expansion.



DRIVERS

- Safer, explainable route guidance
- Dead-zone resilience foundations
- Multilingual alerts + route progress



GOVERNMENT / REGION

- Road-event evidence foundation
- Accessibility intelligence for NER corridors
- Scalable district/state deployment model

NOT JUST A FLEET TRACKER

A resilient decision layer for when roads, weather and connectivity cannot be assumed.

BENEFIT LENS

SOCIAL

Driver safety & accessibility

ECONOMIC

Fewer avoidable disruptions

ENVIRONMENTAL

Reduce needless detours / idling

STRATEGIC

NER-first, pan-India reusable architecture



FLEET MANAGERS

- Live operational visibility
- Route trade-offs + human reroute
- Clear LIVE / STALE / NO CONTACT states



BUSINESS / COMMUNITY

- More reliable deliveries
- Less manual coordination during disruption
- Better route transparency for essential logistics

RESEARCH BASIS / DATA AVAILABILITY



GSI - NATIONAL LANDSLIDE FORECASTING CENTRE

Bhusanket exposes susceptibility maps, forecast bulletins and a field-validated landslide inventory. 36,072 inventory entries were shown on the portal (accessed 02 Sep 2026).



MDoNER - LOGISTICS MODERNISATION ALIGNMENT

MDoNER reports an Industry 4.0 + Logistics Roadmap for NER, approved in May 2025 and under implementation. Our platform directly targets digital logistics decision support.



TECHNICAL RESEARCH

OSRM: alternative routes + turn instructions.
Open-Meteo: coordinate-based forecast variables.
MapLibre + PostGIS: interactive geospatial rendering and spatial persistence.

PRIMARY REFERENCES

1

GSI Bhusanket / NLFC
<https://bhusanket.gsi.gov.in/>

2

MDoNER
<https://www.mdoner.gov.in/>

3

OSRM API Documentation
<https://project-osrm.org/docs/>

4

Open-Meteo Forecast API
<https://open-meteo.com/en/docs>

5

MapLibre GL JS
<https://maplibre.org/maplibre-gl-js/docs/>

6

OpenStreetMap Attribution / Usage
<https://www.openstreetmap.org/copyright>

EVIDENCE POLICY FOR FUTURE ROAD MEMORY

No news ≠ road reopened. CLOSED → REPAIR REPORTED → AWAITING VERIFICATION → VERIFIED OPEN.

Future sources: GSI + BRO / NHIDCL / NHAI / State PWD / district authority + trusted fleet evidence.

ML is introduced only after labelled ground truth and validation.